Class 15: Active Portfolio Management II

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Today's Question

A portfolio manager at your firm got caught trading on inside information. Should you have seen it coming from her returns?

Positioning

Where We've Been	Where We Are	Where We're Going
Measuring α	Skill vs. luck vs. cheating	Performance evaluation frameworks

Learning Objectives

By the end of today's class, you should be able to:

1. Describe how abnormal returns can signal illicit activity
2. Explain the challenges in distinguishing skill from luck
3. Articulate the regulatory framework around insider trading (high level)

Recap: Treynor-Black

From last class: active management as a core/satellite approach.

- One **active** portfolio (satellite) based on alpha views
- One **passive** portfolio (core) — the market index

What does the typical mutual fund represent?

- Depends on the fund benchmark
- Sector/style funds vs. balanced funds

For an undiversified fund, passive investing is done “after-market.”

Extending T-B: The Black-Litterman Model

Treynor-Black addresses the trade-off between diversification and conviction.

- Even ex-ante correct beliefs can be wrong after the fact!
- i.e. alpha belief is right, but error term ε went the wrong way

Black-Litterman extends this by admitting uncertain beliefs about alpha and other inputs *beforehand*.

- Think Bayesian!

Black-Litterman Cookbook

The Black-Litterman approach incorporates three things:

1. Optimal portfolio design (mean-variance trade-off)
2. Active trading (non-zero alpha)
3. Uncertainty (Bayesian shrinkage)

The approach is somewhat technical, so we will walk through the cookbook.

For implementation details: https://python-advanced.quantecon.org/black_litterman.html

Black-Litterman: Step 1

Invert expected returns based on observed portfolio weights and covariances.

- "What must others believe about $E(r)$ to justify these weights?"
- With the universe of stocks, the CAPM already does this

This gives a noisy estimate of expected returns from a market equilibrium.

Assign a level of uncertainty to this estimate (τ).

Black-Litterman: Step 2

Come up with an **independent estimate** of $E(r)$ — your "view."

Assign a level of uncertainty to this estimate (ω).

Examples:

- "I believe NVDA will outperform the market by 2% next quarter" (with some confidence)
- "I believe energy stocks will underperform" (with less confidence)

Black-Litterman: Step 3

Use **Bayes' rule** to formulate the posterior distribution of $E(r)$:

Posterior = weighted average of:

- Equilibrium estimates (weighted by $1/\tau$)
- Your views (weighted by $1/\omega$)

More confident views get **more weight**. Less confident views shrink toward the market equilibrium.

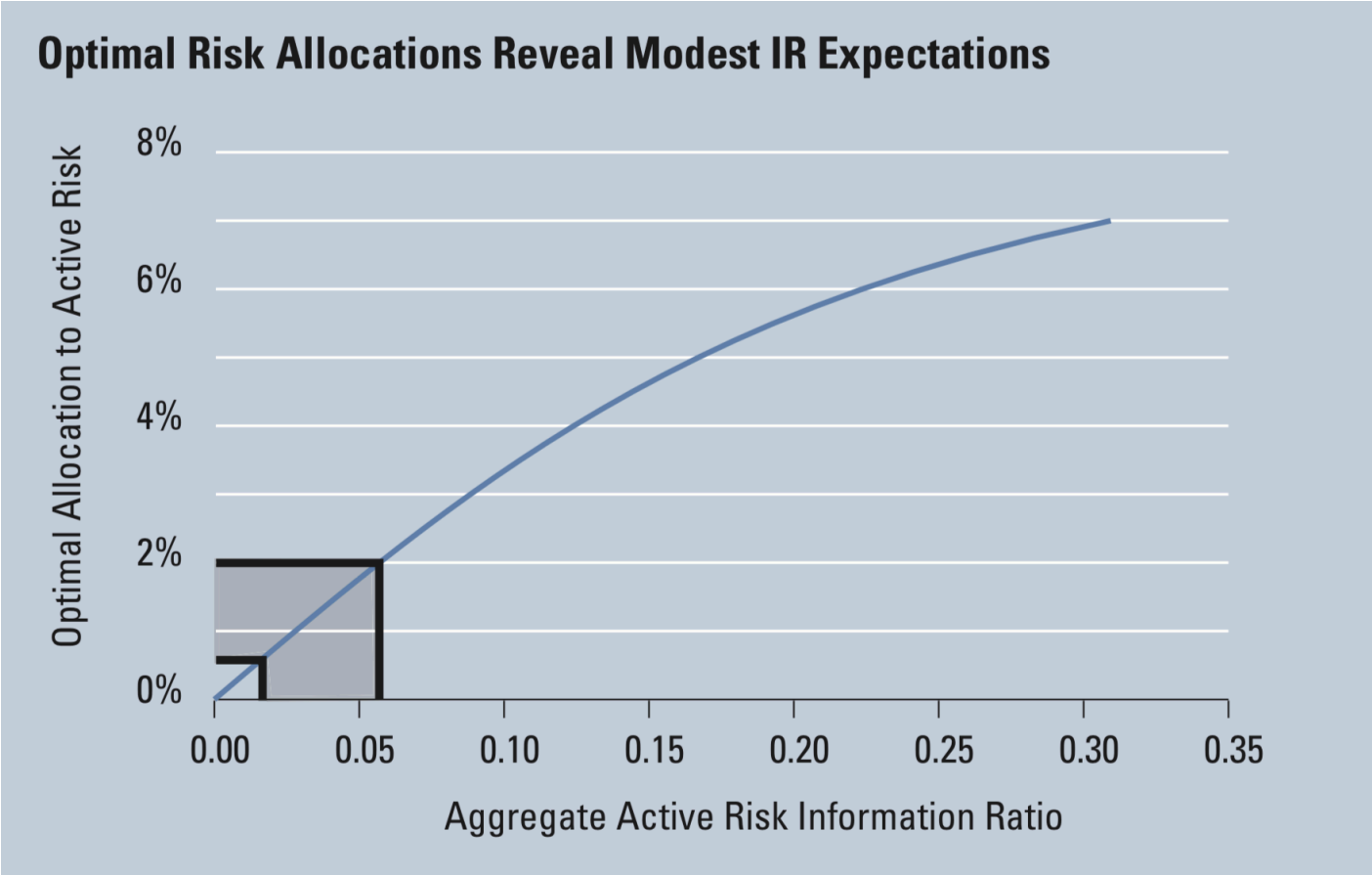
Proceed with mean-variance optimization based on the refined expected returns and covariance matrix.

Does Active Management Work?

We now have the theory (T-B, B-L). But does active management actually deliver alpha in practice?

Are funds over-indexing? Bob Litterman says so...

Does Active Management Work?



Fund Alphas: Net of Expenses

Fama and French estimate the distribution of fund alphas after costs:

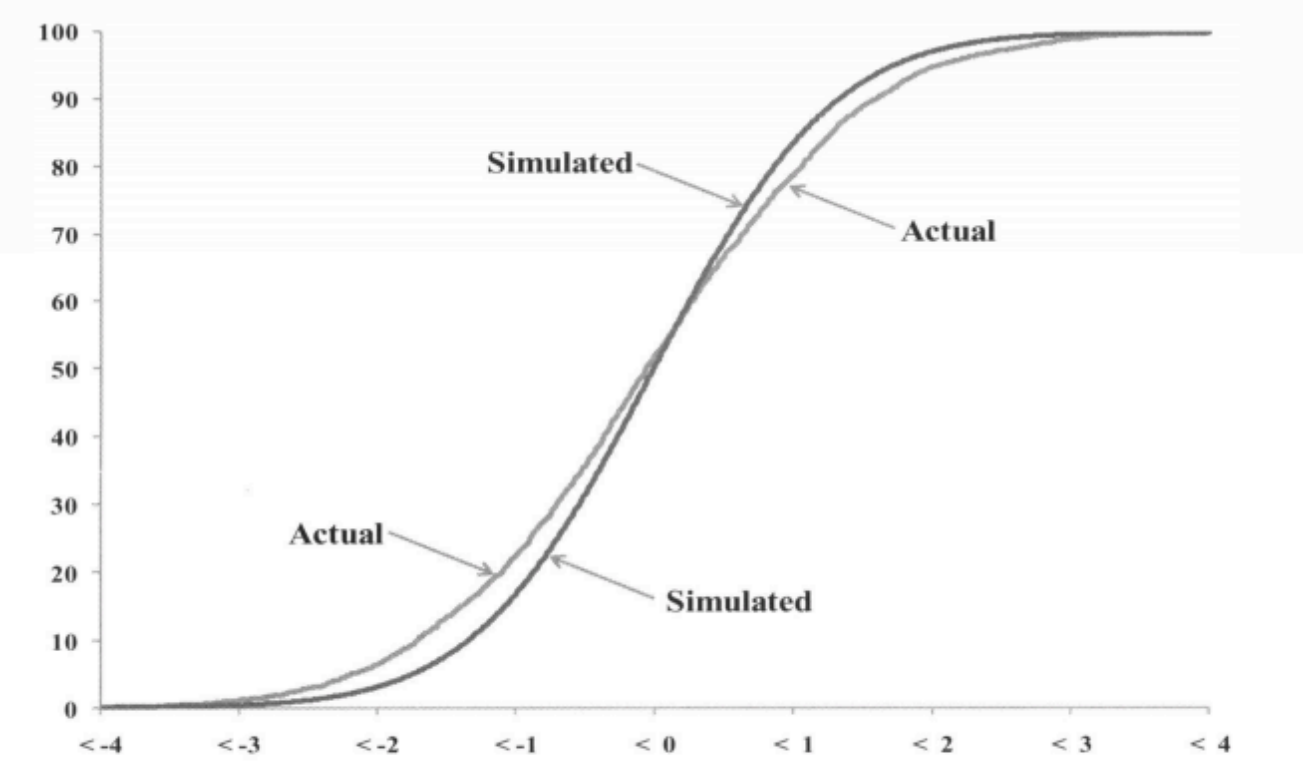


Figure 2. Simulated and actual cumulative density function of three-factor $t(\alpha)$ for gross returns, 1984–2006.

Fund Alphas: Gross of Expenses

Before costs, the picture is less grim — but still centered near zero:

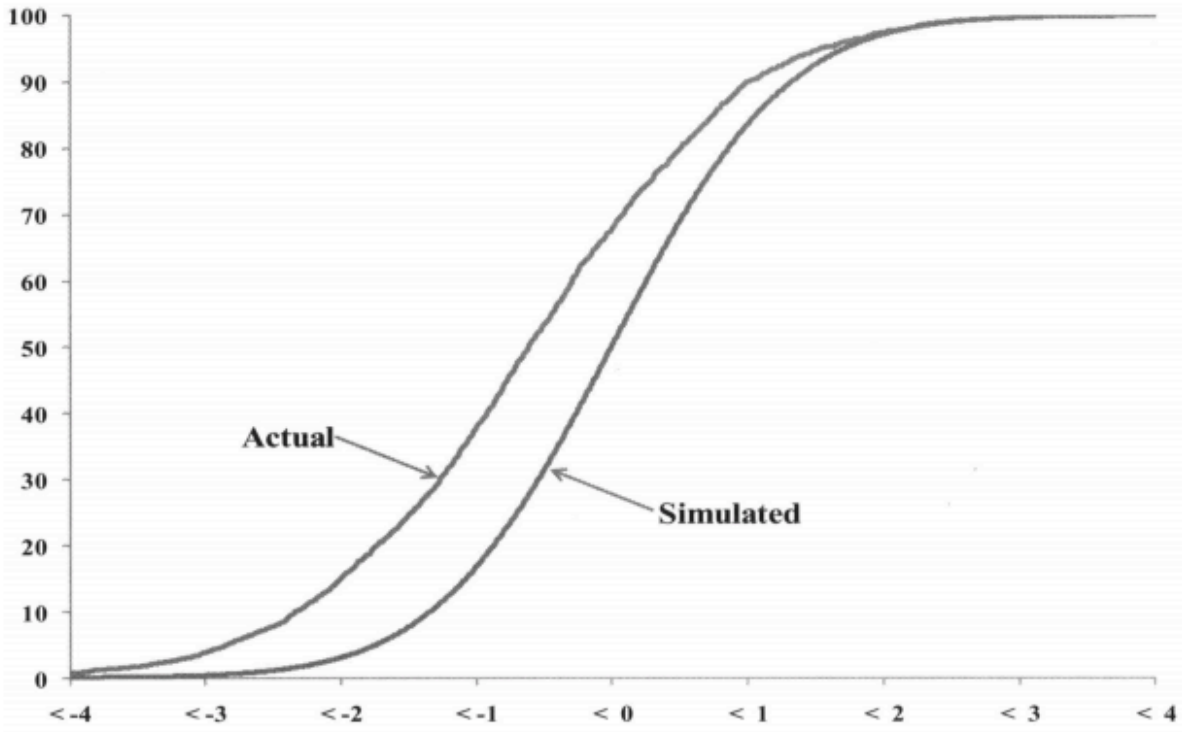


Figure 1. Simulated and actual cumulative density function of three-factor $t(\alpha)$ for net returns, 1984–2006.

Hot Hands?

Evidence for persistent superior performance is **weak**.

Malkiel (1995) tracks funds based on above/below median performance:

Initial Year		Next Year: Winner	Next Year: Loser	Percent Repeat Winners
1971–1979	Winner	682	365	65.1
	Loser	371	675	
1980–1990	Winner	1189	1111	51.7
	Loser	1087	1203	

Is alpha the right measure? Van Binsbergen and Berk (2011)

Maybe We're Measuring 'Active' Wrong

The persistence results lump together truly active managers and “closet indexers” charging active fees.

How do we observe if funds are *actually* active?

- One idea: tracking error — $\mathrm{sd}(r_{\text{fund}} - r_{\text{index}})$

Alternatively, the **active share** (Cremers and Petajisto, 2009):

$$\text{Portfolio} = \text{Index} + (\text{Portfolio} - \text{Index})$$

Distinguish the truly active funds from those who only *advertise* themselves as active.

Active Share

Active Share quantifies active portfolio holdings — compare fund weights to benchmark:

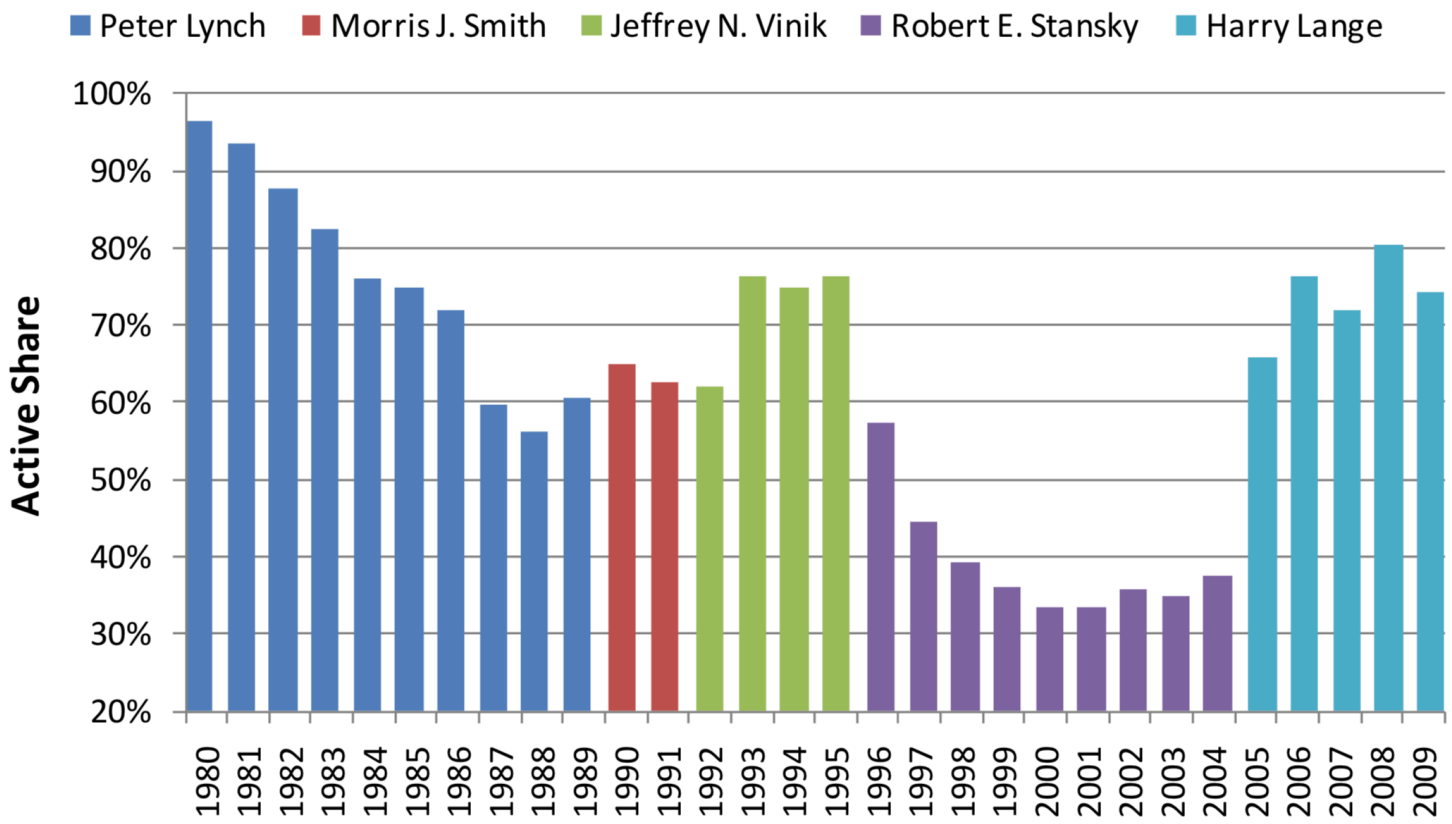
$$\text{Active Share}_{\text{fund}} = \frac{1}{2} \sum \left| w_{\text{fund, stock}} - w_{\text{index, stock}} \right|$$

- Indicates the size of active positions as a fraction of the entire portfolio
- Always between 0% and 100% for mutual funds
  - e.g. 34% for Fidelity Magellan at the end of 2003

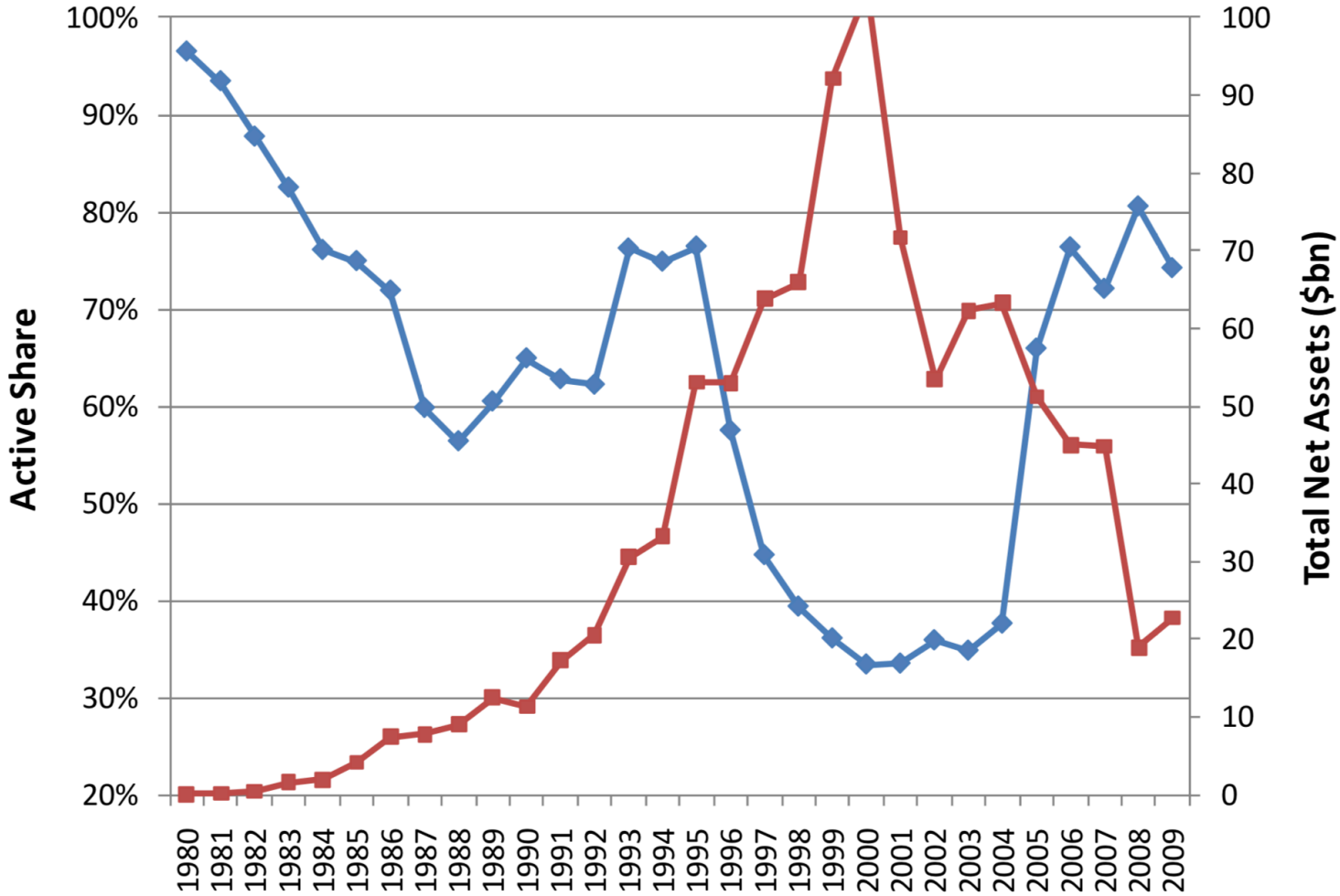
Case Study: Fidelity Magellan

Magellan was legendary under Peter Lynch. By 2003, its active share had collapsed:

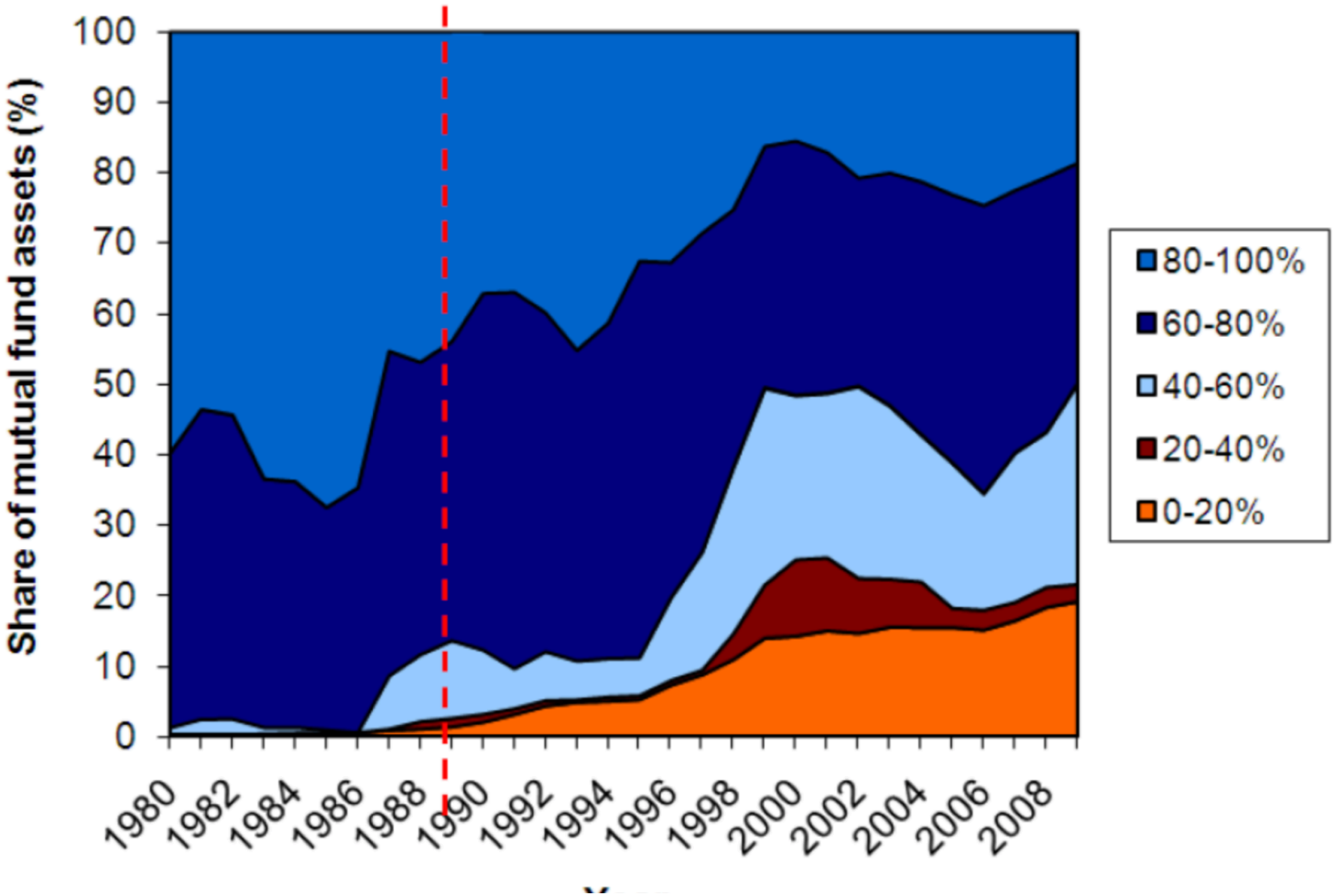
Case Study: Fidelity Magellan



Magellan: Closet Indexing in Detail

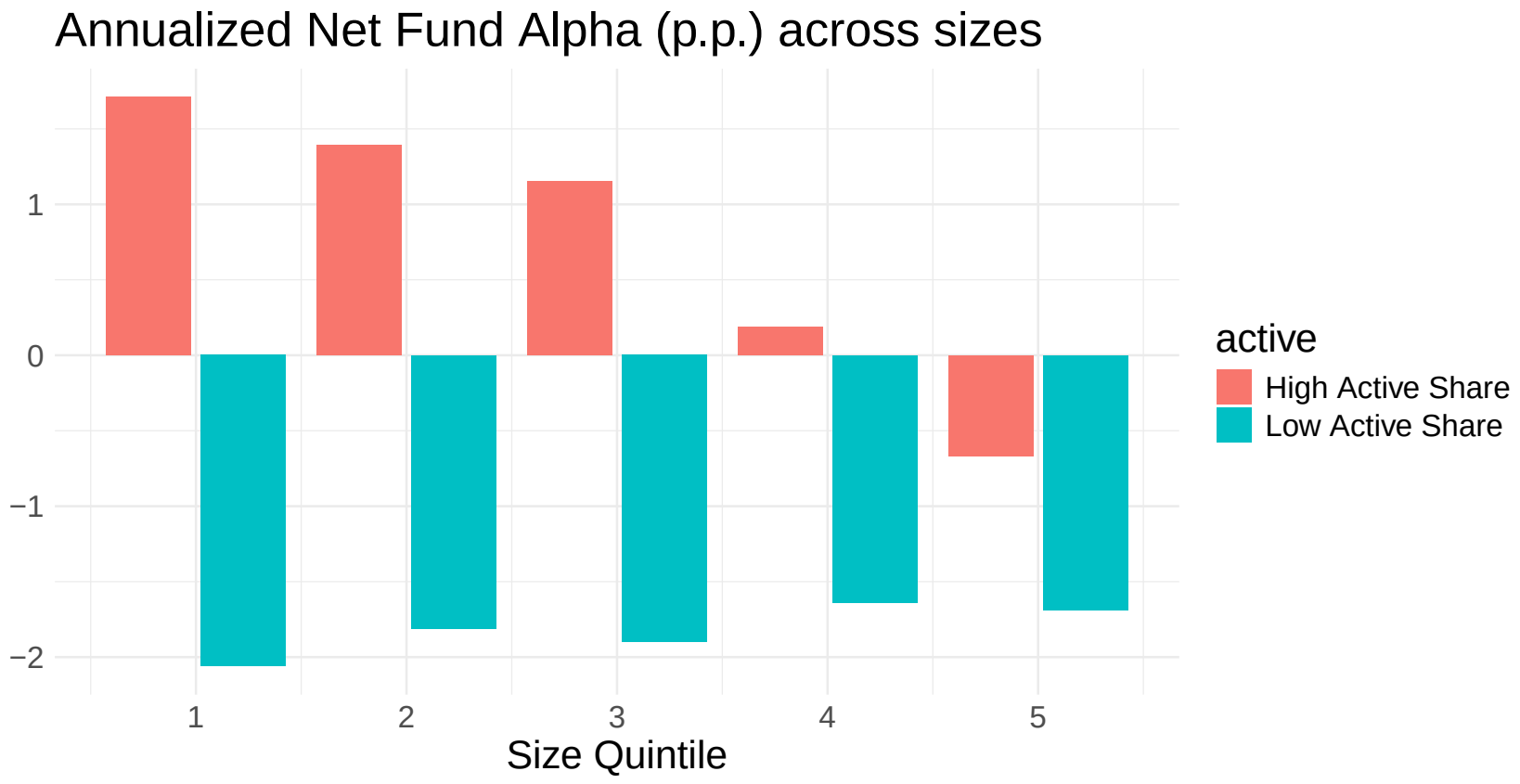


The Industry Trend: Declining Active Share

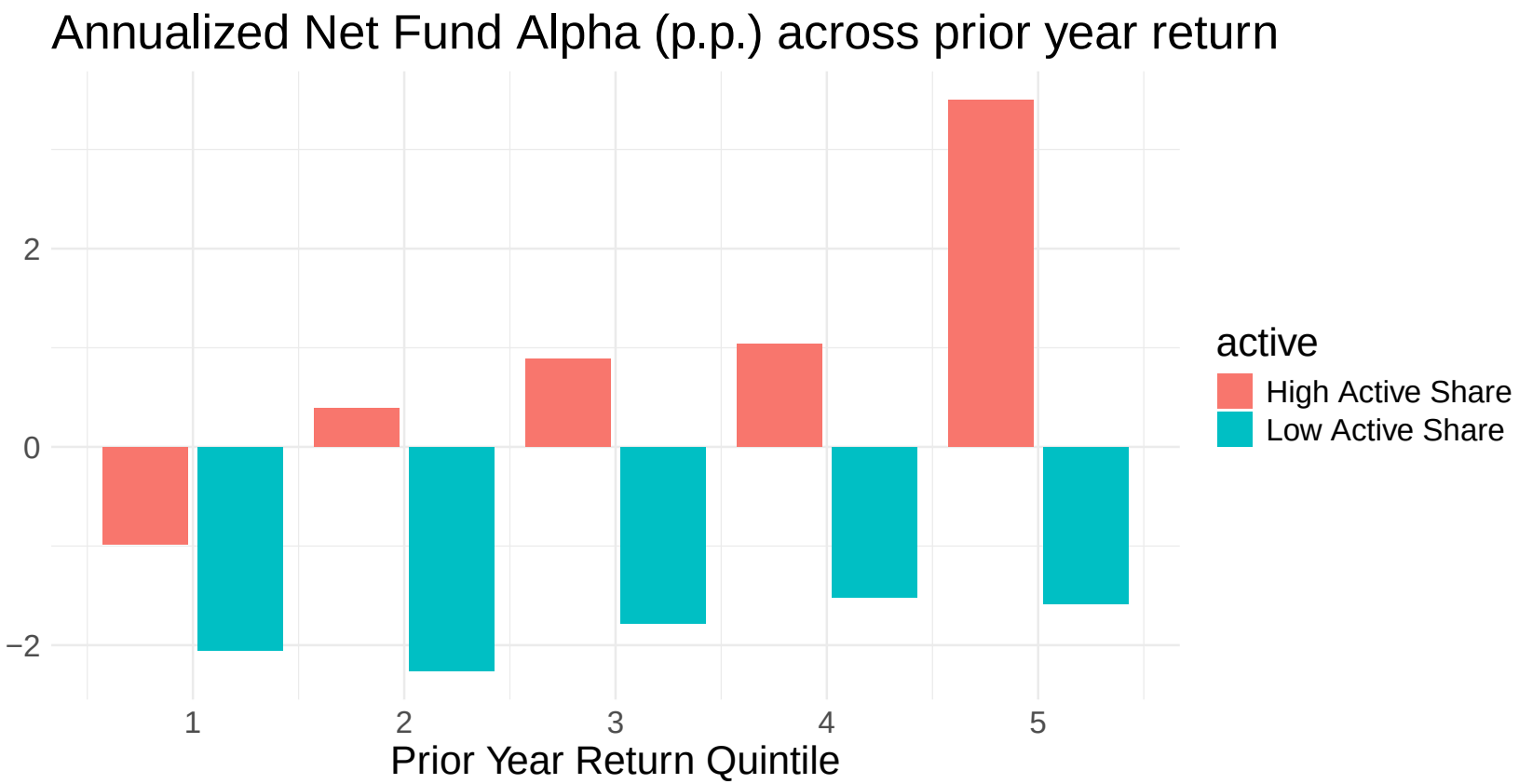


High Active Share Funds Outperform

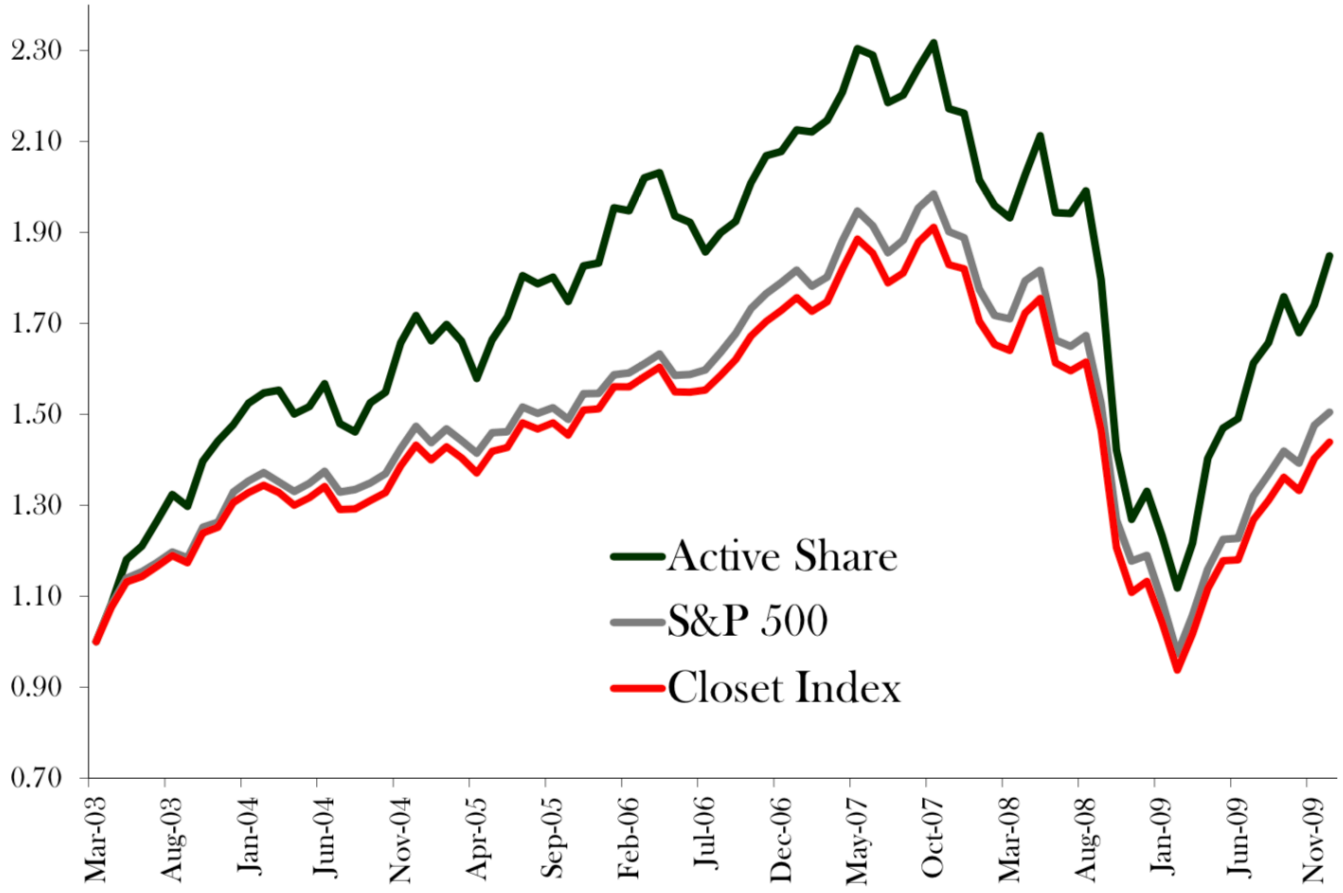
Sorting funds by active share *and* size reveals a clear pattern:



High Active Share Funds Show More Persistence



Active Share and Performance



So Some Managers Do Generate Alpha...

The active share evidence suggests that *truly* active managers can outperform.

But when we observe abnormal returns, three possible explanations remain:

1. **Skill:** Superior analysis and information processing
2. **Luck:** Random favorable outcomes
3. **Cheating:** Insider trading or other illicit activity

How do we tell them apart?

Detecting Abnormal Returns

How might insider trading show up in a manager's track record?

- Consistently positive alpha that is **too good to be true**
- Unusually high returns around corporate events (mergers, earnings)
- Returns that predict future news releases (e.g., consistently buying before mergers are announced)
- Low correlation with known risk factors (alpha without factor exposure)

The SEC uses statistical methods to flag suspicious trading patterns — but distinguishing skill from cheating is fundamentally difficult.

The Regulatory Framework

Insider trading (high-level):

- Illegal to trade on material, non-public information (MNPI)
- Applies to corporate insiders *and* anyone who receives tips
- SEC enforcement relies on pattern detection, whistleblowers, and electronic surveillance

Key challenge: *proving* that returns came from MNPI rather than skill.

Connection to our framework:

- α from skill looks the same as α from insider information
- You need *context* beyond returns to distinguish them

Distinguishing Skill from Luck

Even without cheating, how many years of data do you need to tell if a manager is skilled?

Suppose a manager has $\mathrm{IR} = 0.5$ (a *very* good information ratio).

- Standard error of alpha estimate $\approx \sigma(\varepsilon)/\sqrt{T}$
- You need roughly **16 years** of data to reject $\alpha = 0$ at the 5% level!

With $\mathrm{IR} = 0.25$ (more realistic): **64 years**.

Implication: most track records are too short to distinguish skill from luck with any confidence.

Back to Today's Question

Should you have seen it coming from her returns?

Probably not:

- Abnormal returns alone are ambiguous: skill, luck, or fraud
- Statistical detection requires **very long** track records
- Context matters: *when* do the abnormal returns cluster? Around events = suspicious
- Regulatory enforcement supplements (but cannot replace) statistical analysis

The tools we've built (IR, alpha decomposition, active share) help us *ask* the right questions — but they rarely give definitive answers.

Next Class: March 31

Topics: Portfolio Evaluation

- Time-weighted vs. dollar-weighted returns
- Performance attribution: allocation vs. selection
- Sharpe, Treynor, Jensen's α, Information Ratio